

ODMA–URA research record

## The explicit factorised URA framework

From the unrestricted  $n \times 2^B$  codebook to implicit operators, learned D0/D1, and controlled structure

|         |                                                             |
|---------|-------------------------------------------------------------|
| Stage   | 3 of 5                                                      |
| Updated | 5 September 2026                                            |
| Status  | Exact small- $B$ framework validated; jobs 018–022 complete |

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

## 1 Research principle and starting point

The support-oracle experiment suggested that the principal loss was not merely energy sparsity. The next aim was to search for structured codebooks without deciding in advance that one named scheme—ODMA, CCS, SPARC, or another—was the answer. The correct unconstrained reference is the full common codebook

$$\Phi = [\phi_0, \dots, \phi_{M-1}] \in \mathbb{C}^{n \times M}, \quad M = 2^B,$$

and the full count vector

$$a = (a_0, \dots, a_{M-1})^T \in \mathbb{Z}_+^M, \text{quad}y = \Phi a + z.$$

Every column is one complete  $B$ -bit message. At  $B = 12$ ,  $M = 4096$ ; at  $B = 14$ ,  $M = 16384$ ; at  $B = 100$ , the explicit matrix and even the length- $M$  vector are impossible to store. Nevertheless, the explicit model is the right algebraic starting point because it defines the broad model class before computational restrictions are imposed.

The research philosophy is to remove exponential storage or search with the weakest structural restriction that works, and to measure each restriction against the broad explicit class. A compact representation is useful only if it retains good codebook geometry and admits complete-message recovery.

## 2 Layer 1: latent contribution decomposition

Suppose a complete codeword is the sum of  $L$  contributions. Component  $\ell$  has a local alphabet of  $N_\ell$  full-frame atoms

$$\Phi_\ell = [\phi_{\ell,1}, \dots, \phi_{\ell,N_\ell}] \in \mathbb{C}^{n \times N_\ell}.$$

Global message  $m$  chooses index  $i_\ell(m) \in \{1, \dots, N_\ell\}$ . Define the one-hot selection matrix

$$T_\ell \in \{0, 1\}^{N_\ell \times M}, \quad T_\ell[:, m] = e_{i_\ell(m)}.$$

Then the complete waveform and codebook are

$$\phi_m = \sum_{\ell=1}^L \phi_{\ell, i_\ell(m)}, \quad \boxed{\Phi = \sum_{\ell=1}^L \Phi_\ell T_\ell.}$$

This does not require the message to be semantically split. The fully general one-shot codebook is the special case

$$L = 1, \quad \Phi_1 = \Phi, \quad T_1 = I_M.$$

For global counts  $a$ , define local aggregate counts

$$s_\ell = T_\ell a \in \mathbb{Z}_+^{N_\ell}.$$

Then

$$y = \Phi a + z = \sum_{\ell=1}^L \Phi_\ell s_\ell + z.$$

Each  $s_\ell$  is many-to-one: it records how many active messages used each local atom, not which atoms across different components belonged to the same complete message. This observation later becomes the association problem.

### 3 Layer 2: product structure within a component

Component  $\ell$  may itself be generated by resource operators and local codewords. Let

$$R_{\ell,q} \in \mathbb{C}^{n \times d_\ell}, \quad q = 1, \dots, Q_\ell, \quad C_\ell = [c_{\ell,1}, \dots, c_{\ell,V_\ell}] \in \mathbb{C}^{d_\ell \times V_\ell}.$$

For a fixed operator  $q$ ,  $R_{\ell,q}C_\ell$  contains all  $V_\ell$  codewords passed through that operator. The complete product library is

$$B_\ell = [R_{\ell,1}C_\ell \mid \dots \mid R_{\ell,Q_\ell}C_\ell] \in \mathbb{C}^{n \times Q_\ell V_\ell}.$$

Not every pair  $(q, v)$  must be legal. Let

$$U_\ell \in \{0, 1\}^{Q_\ell V_\ell \times N_\ell}$$

select  $N_\ell$  valid local atoms, with one nonzero per column for an exact selection. Thus

$$\Phi_\ell = B_\ell U_\ell.$$

Substitution gives the central framework:

$$\Phi = \sum_{\ell=1}^L [R_{\ell,1}C_\ell \mid \dots \mid R_{\ell,Q_\ell}C_\ell] U_\ell T_\ell.$$

The factors have different meanings and should not be conflated:

| Object and shape                                   | Meaning                                  | Computational or modelling implication                                     |
|----------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------|
| $L$                                                | Number of additive contributions         | $L > 1$ creates a path/association problem unless the tuple is constrained |
| Operator $R_{\ell,q}$<br>$n \times d_\ell$         | Placement, mask, spreading, or transform | Controls resource occupation and geometry                                  |
| Alphabet $C_\ell$<br>$d_\ell \times V_\ell$        | Local value/codeword alphabet            | Natural continuous learning target                                         |
| Validity $U_\ell$<br>$Q_\ell V_\ell \times N_\ell$ | Legal local $(q, v)$ pairs               | Usually a fixed discrete map                                               |
| Mapping $T_\ell$<br>$N_\ell \times M$              | Complete message to local atom           | Contains global semantics and is exponential if stored arbitrarily         |

## 4 Representation library

The value of the factorisation is that named constructions become controlled restrictions of one global model.

### 4.1 Dense one-shot codebook

For arbitrary  $\Phi_{\text{dense}} \in \mathbb{C}^{n \times M}$ , take

$$L = 1, \quad Q_1 = 1, \quad d_1 = n, \quad V_1 = N_1 = M,$$

$$R_{1,1} = I_n, \quad C_1 = \Phi_{\text{dense}}, \quad U_1 = T_1 = I_M.$$

No model-class restriction has been introduced.

## 4.2 Sparse-global codebook

Use the same one-component representation but constrain each column of  $C_1$ :

$$\|C_1[:, m]\|_0 = s, \text{quad}\|C_1[:, m]\|_2 = 1.$$

Supports may be independent across messages. This preserves a broad family of  $\binom{n}{s}$  masks and became the most promising explicit control.

## 4.3 ODMA/product construction

Let each  $R_q$  place or mask a shared local codebook  $C$ . With all pairs legal,

$$\Phi = [R_1 C \mid \cdots \mid R_Q C], \quad L = 1, \quad U = T = I_{QV}, \quad M = QV.$$

ODMA is a sparse placement special case. Sharing only  $Q$  supports across  $M$  messages is a much stronger restriction than giving every message its own sparse support.

## 4.4 Restricted pattern-codeword pairs

If only a subset of  $(q, v)$  pairs is valid,  $U$  selects them:

$$\Phi = [R_1 C \mid \cdots \mid R_Q C] U, \text{quad} N = M, \text{quad} T = I_M.$$

This separates the geometry of the available product library from the combinatorial selection of messages.

## 4.5 Sectioned CCS or SPARC-style construction

Let  $m$  map to one local index  $i_\ell(m)$  in each section. Then

$$\phi_m = \sum_{\ell=1}^L \Phi_\ell e_{i_\ell(m)}.$$

Disjoint-section CCS uses  $R_{\ell,1}$  to embed a section dictionary into its own channel-use interval. A SPARC-like representation may instead use  $R_{\ell,1} = I_n$  and  $C_\ell = \sqrt{E_\ell} A_\ell$ . An outer code constrains which tuples  $(i_1, \dots, i_L)$  are legal; algebraically that constraint is carried by the collection of  $T_\ell$  maps.

## 4.6 Spreading, interleaving, and hybrid families

Dense diagonal signatures, permutations, interleavers, slot embeddings, and learned linear maps are all possible  $R_{\ell,q}$ . A hybrid may use several components, each with its own operator bank and local alphabet. The factorisation is a representation of transmitted waveforms; it does not imply that conventional decoders for all these schemes are the same.

## 5 Explicit versus implicit implementation

The non-scalable mathematical object is the matrix  $\Phi \in \mathbb{C}^{n \times M}$ . The implementation avoids materialising it by storing:

- the continuous  $R$  and  $C$  factors or compact operator parameters;
- integer arrays that identify each legal local atom, replacing dense one-hot  $U$ ;
- a length- $M$  `msg_to_atom` lookup, replacing dense one-hot  $T$ .

For a vector  $v \in \mathbb{C}^M$ , the forward operator is

$$\Phi v = \sum_{\ell} B_{\ell} U_{\ell} (T_{\ell} v).$$

$T_{\ell} v$  aggregates global coefficients by local atom;  $U_{\ell}$  scatters them into the  $(q, v)$  product library;  $C_{\ell}$  forms local vectors; and  $R_{\ell, q}$  maps them to the resource grid. The exact adjoint reverses the chain:

$$\Phi^H r = \sum_{\ell} T_{\ell}^H U_{\ell}^H B_{\ell}^H r.$$

This reduces storage from  $O(nM)$  to the factor storage plus  $O(M)$  integer mappings and decoder state. It is valuable at  $B = 12$  or  $14$ , but it is not a  $B = 100$  solution because both  $T_{\ell}$  and the output  $\Phi^H r \in \mathbb{C}^M$  retain the global message axis.

## 6 Gradient flow through the factorisation

Let a batch of global counts be  $A \in \mathbb{R}^{P \times M}$  and let the encoder output be  $Y = A\Phi^T$ . For loss  $\mathcal{L}$  with upstream gradient  $G_Y = \partial\mathcal{L}/\partial Y$ , the count gradient is the adjoint operation

$$\frac{\partial\mathcal{L}}{\partial A} = G_Y \Phi^*.$$

For one component and operator, write the aggregated coefficients as  $X_{\ell, q}$  and

$$Z_{\ell, q} = X_{\ell, q} C_{\ell}^T, \text{quad} Y_{\ell} = \sum_q Z_{\ell, q} R_{\ell, q}^T.$$

Then, suppressing conjugation conventions,

$$\begin{aligned} \frac{\partial\mathcal{L}}{\partial Z_{\ell, q}} &= G_Y R_{\ell, q}^*, \\ \frac{\partial\mathcal{L}}{\partial C_{\ell}} &= \sum_q \left( \frac{\partial\mathcal{L}}{\partial Z_{\ell, q}} \right)^T X_{\ell, q}, \quad \frac{\partial\mathcal{L}}{\partial R_{\ell, q}} = G_Y^T Z_{\ell, q}^*. \end{aligned}$$

Autodifferentiation implements these adjoint chains without building  $\Phi$ . Continuous  $C$  and  $R$  are learnable. Exact integer  $U$  and  $T$  have no ordinary gradient; changing them requires discrete design, search, or an explicitly labelled relaxation.

## 7 Feasibility constraints and learning objective

The broad factorisation is not a proposal to learn everything simultaneously. The communication problem is

$$\min_{\theta, \gamma} \mathbb{E}[\mathcal{L}(a, D_{\gamma}(y; \Phi_{\theta}))] \quad \text{s.t.} \quad \|\phi_m(\theta)\|_2^2 = 1 \quad \forall m,$$

with final evaluation in PUPE or required  $E_b/N_0$ , not only surrogate loss.

Important hard conditions are:

1. equal unit energy for every global codeword, enforced by parameterisation or projection;

2. injectivity of  $m \mapsto (i_1(m), \dots, i_L(m))$  unless deliberate ambiguity is being tested;
3. exact support budgets for sparse families rather than hoping an  $\ell_1$  penalty produces them;
4. balanced resource use and controlled intersections as diagnostics or construction constraints;
5. fixed  $U, T$  in early experiments, so continuous learning cannot silently change message semantics.

The first surrogate combined balanced support BCE with a count Smooth-L1 term. Count MSE,

$$\mathcal{L}_{\text{count}} = M^{-1} \|\hat{a} - a\|_2^2,$$

is another simple option, but no differentiable surrogate should be confused with PUPE.

## 8 D0: exact data consistency with learned scalar denoising

For batch size  $P$ , initialise  $a^0 = 0 \in \mathbb{R}^{P \times M}$ . Layer  $t$  performs

$$r^t = y - \Phi a^t, qquad g^t = \Re\{\Phi^H r^t\}, qquad u^t = a^t + \eta_t g^t.$$

With known  $K_a$ , a Bernoulli activity approximation uses  $\rho = K_a/M$ . D0 converts each coordinate to a support logit

$$\lambda_j^t = \text{clip} \left[ \gamma_t \frac{u_j^t - 1/2}{\tau_t} + \beta_t \log \frac{\rho}{1 - \rho} + b_t, -30, 30 \right].$$

The soft proposal is proportional to  $\sigma(\lambda^t)$  and is normalised to total mass  $K_a$ ; damping produces  $a^{t+1}$ . Hard nonnegative integer projection occurs only at evaluation.

The physical forward/adjoint pair is exact. Learned scalars calibrate step size, evidence scale, prior, temperature, bias, and damping. Gradients through all unrolled layers reach both decoder parameters and continuous encoder factors. D0 still computes an  $M$ -length vector.

## 9 D1: learned nonlocal factor context

D1 starts with the same  $r^t, g^t, u^t$  and base logits. For an  $L = 1$  product bank, message  $j$  has labels  $(q_j, v_j)$ . Its features include  $u_j^t, a_j^t, g_j^t$ , base logit, pattern mass, residual/noise scales, and prior logit. Latent slot attention pools context:

- among messages sharing the same pattern  $q$ ;
- across patterns for a shared local value  $v$ ;
- globally across the message set.

A small network adds correction  $\Delta_j^t$  to the D0 logit. The correction head is zero-initialised, so D1 initially equals D0 exactly. For fixed slot count, the operation is linear rather than quadratic in  $M$ , but it remains indexed by all  $M$  messages and is several times slower.

The factor-aware interpretation must be made carefully: D1 also improves dense controls, so a D1 gain alone does not prove that product structure was exploited. A causal factor benefit requires matched controls or ablations.

## 10 Validation before performance

The implementation checked:

- implicit versus materialised forward maps and adjoint inner products;
- selected-column construction;

- exact representation of legacy ODMA and reduced sectioned CCS constructions;
- tiny exact count ML, including message collisions;
- normal  $B = 14$ ,  $M = 16384$  training without constructing an  $n \times M$  matrix.

These certify implementation of the intended algebra at materialisable  $B$ . They do not prove paper-faithful AMP state evolution or large- $B$  complete-message decoding.

## 11 Jobs 021–022 and the promising sparse-global result

The controlled bank fixed  $L = 1$ ,  $U = T = I$ , and  $d = n$ . It compared dense, shared product, independently sparse-global, and four-mask ODMA encoders. The four geometries were  $(B, n) = (12, 128)$ ,  $(12, 256)$ ,  $(14, 128)$ , and  $(14, 256)$ , under common D0/D1 training and evaluation.

The defensible conclusions are:

1. D0 improved only slightly over its matched-filter calibration, about 0.004 PUPE on average.
2. D1 improved high-SNR PUPE by about 0.042 on average, but did so for dense controls too; a uniquely factor-aware gain was not isolated. Median runtime was about 5.2 times D0.
3. Product sharing was competitive but did not beat dense controls on average; learning  $C$  had mixed effect.
4. Independent 25%-sparse global supports stayed within 0.016 PUPE of dense in all four geometries and were essentially tied at  $n = 256$ .
5. Four-mask ODMA was much worse and left roughly 30% of rows unused. Equal nonzero fraction therefore did not imply equal recovery geometry.

**Evidence status:** Measured jobs 021–022. The sparse-global result is an explicit  $M = 2^B$  existence result under the tested learned decoders; it is not yet a compact generator or scalable receiver.

The original staged programme was therefore sensible: dense  $\rightarrow$  sparse global  $\rightarrow$  product/ODMA restrictions  $\rightarrow$  valid-atom restrictions  $\rightarrow L > 1 \rightarrow$  mapping design. The next implementation attempted to remove the remaining  $M$  axis by moving to local section states. Report 4 explains why that execution succeeded algebraically but the chosen outer association and decoder did not preserve the strong  $L = 1$  landscape.